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COOPERATIVE AND OPTIMIZATION-BASED FRAMEWORK FOR AERIAL MANIPULATION

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ABSTRACT

CO-FAST presents a novel ground-braking framework for cooperative aerial manipulation based on swarms of unmanned aerial vehicles connected to a shared payload through motorized, extensible cables. The system is modeled as a floating-base parallel kinematic structure [1], enabling full six-degree-of-freedom payload manipulation while significantly simplifying swarm formation control. By employing independently actuated winch-driven cables[2], CO-FAST decouples payload motion from the geometric configuration of the aerial swarm, overcoming key limitations of traditional fixed-cable approaches that rely solely on robot positioning.

This decoupling can be exploited within a hierarchical optimization-based control architecture [3] that balances stability, energy efficiency, and manipulation capability [4]. Unlike quasi-static methods, the proposed framework can explicitly account for dynamic couplings and external disturbances, enabling responsive and robust control.

The capabilities of CO-FAST are demonstrated by two high-impact application scenarios: cooperative aerial wildfire suppression, leveraging game-theoretic resource allocation strategies [5], and precise aerial transportation [6] and assembly of heavy or asymmetric loads in construction environments. In these contexts, the proposed approach shows the potential to outperform conventional ground-based solutions in terms of accessibility, deployment time, and operational cost. By combining the maneuverability of aerial platforms with the precision of cable-driven parallel robotics, CO-FAST advances the state of the art in distributed aerial manipulation and contributes to the development of sustainable, flexible solutions for emergency response and infrastructure operations.

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